The documentation and process conversion measures necessary to comply with this revision shall be completed by 25 July 2025.

INCH-POUND

MIL-PRF-19500/385K
w/AMENDMENT 2
25 March 2025
SUPERSEDING
MIL-PRF-19500/385K
w/AMENDMENT 1
14 February 2020

PERFORMANCE SPECIFICATION SHEET

# TRANSISTOR, JUNCTION FIELD EFFECT, N-CHANNEL, RADIATION HARDENED, SILICON, DEVICE TYPES 2N4856 THROUGH 2N4861, JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

## 1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for N-channel, depletion mode, silicon, junction field-effect transistors (JFETs). Four levels of product assurance are provided for each device (JAN, JANTX, JANTXV, and JANS). RHA level designators "M", "D", "P", "L", "R", "F", "G" and "H" are appended to the device prefix to identify devices, which have passed RHA requirements.

1.2 Package outlines. The device package outlines are as follows: TO-18 (no suffix) in accordance with figure 1 and surface mount (UB suffix) in accordance with figure 2 for all encapsulated device types.

1.3 Maximum ratings. $T_A$ = +25°C, unless otherwise specified. (1)

| $P_T$ (2) $T_A$ = +25°C | $P_T$ (3) $T_C$ = +25°C | $V_{DS}$, $V_{DG}$ | | $V_{GS}$ | | $I_G$ | $R_{\theta JA}$ | $R_{\theta JC}$ | $T_J$ and $T_{STG}$ |
|---|---|---|---|---|---|---|---|---|---|
| | | 2N4856<br>2N4857<br>2N4858 | 2N4859<br>2N4860<br>2N4861 | 2N4856<br>2N4857<br>2N4858 | 2N4859<br>2N4860<br>2N4861 | | | | |
| W | W | V dc | V dc | V dc | V dc | mA dc | °C/W | °C/mW | °C |
| 0.36<br>0.40, all UB (4) | 1.8 | 40 | 30 | -40 | -30 | 50 | 486<br>325 | 0.097 | -65 to +200 |

(1) These characteristics applicable to all package styles, unless otherwise noted.
(2) Derate linearly 2.06 mW/°C for $T_A$ > +25°C.
(3) Derate linearly 10.3 mW/°C for $T_C$ > +25°C.
(4) Derate linearly 3.08 mW/°C above $T_C$ = +70°C.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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FSC 5961

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* 1.4 Primary electrical characteristics. $T_A$ = +25°C, unless otherwise specified. (1)

| | $I_{DSS}$ (2) $V_{DS}$ = 15 V dc $V_{GS}$ = 0 | | | $V_{DS}$(on) $V_{GS}$ = 0 $I_D$ = 20 mA dc | $V_{GS}$ = 0 $I_D$ = 10 mA dc | $V_{GS}$ = 0 $I_D$ = 5 mA dc | $V_{GS}$(off) $V_{DS}$ = 15 V dc $I_D$ = 0.5 nA dc | | |
|---|---|---|---|---|---|---|---|---|---|
| | 2N4856 2N4859 | 2N4857 2N4860 | 2N4858 2N4861 | 2N4856 2N4859 | 2N4857 2N4860 | 2N4858 2N4861 | 2N4856 2N4859 | 2N4857 2N4860 | 2N4858 2N4861 |
| | mA dc | mA dc | mA dc | V dc | V dc | V dc | V dc | V dc | V dc |
| Min | 50 | 20 | 8 | | | | -4 | -2 | -0.8 |
| Max | 175 | 100 | 80 | 0.75 | 0.50 | 0.50 | -10 | -6 | -4 |

| | $r_{ds}$(on) $V_{GS}$ = 0; $I_D$ = 1.0 mA dc $I_d$ = 100 µA ac(rms), f = 1 kHz | | |
|---|---|---|---|
| | 2N4856 2N4859 | 2N4857 2N4860 | 2N4858 2N4861 |
| | Ω | Ω | Ω |
| Min | | | |
| Max | 25 | 40 | 60 |

(1) These characteristics applicable to all package styles.
(2) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV" and "JANS".

1.5.2 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for this specification sheet from lowest to highest are as follows: "M", "D", "P", "L", "R", "F", "G", and "H".

1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "4856", "4857", "4858", "4859", "4860", and "4861".

1.5.4 Suffix letters. No suffix is used on devices that are packaged in the TO-18 package of figure 1. The suffix letters "UB" are used on devices that are packaged in the surface mount package of figure 2.

1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

* 1.6 Radiation features. The following radiation features are applicable for RHA devices supplied to this specification sheet.

* 1.6.1 Maximum total ionizing dose (TID). The maximum TID that RHA devices were tested to in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-STD-750 are as follows:

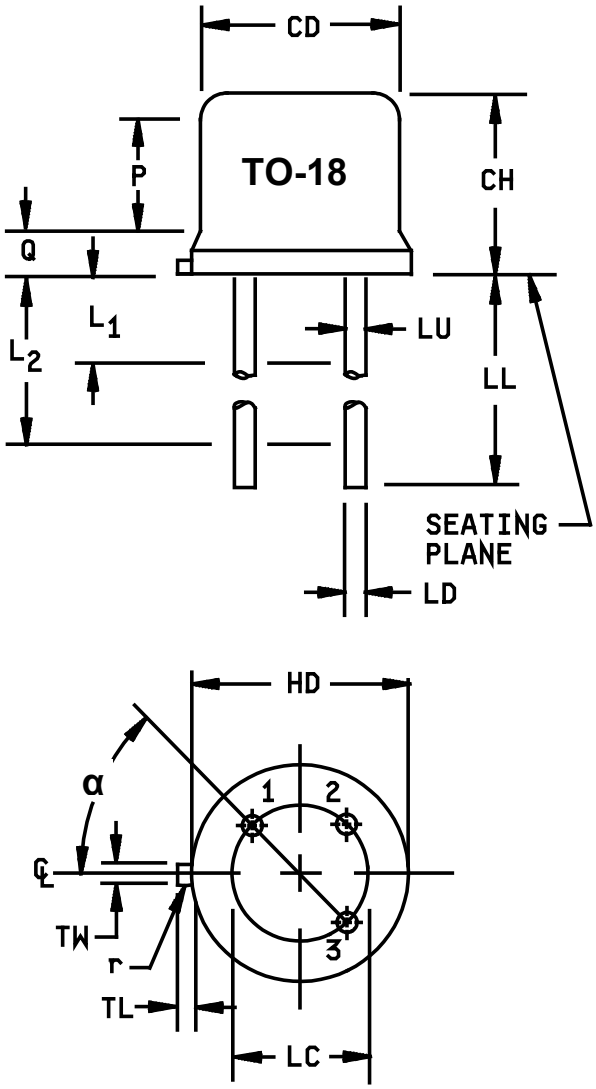
For device type 2N4856 THROUGH 2N4861: ..... 100 krad(Si) 1/

* 1/ The manufacturers supplying these device types have performed characterization testing in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-STD-750. The radiation end point limits are guaranteed only to a maximum TID level of 100 krad(Si).



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| Symbol | Dimensions | | | | Note |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | .178 | .195 | 4.52 | 4.95 | 5 |
| CH | .170 | .210 | 4.32 | 5.33 | |
| HD | .209 | .230 | 5.31 | 5.84 | 4, 5 |
| LC | .100 TP | | 2.54 TP | | 6 |
| LD | .016 | .021 | 0.41 | 0.53 | 7,8 |
| LL | .500 | .750 | 12.70 | 19.05 | 7,8 |
| LU | .016 | .019 | 0.41 | 0.48 | 7,8 |
| $L_1$ | | .050 | | 1.27 | 7,8 |
| $L_2$ | .250 | | 6.35 | | 7,8 |
| P | .100 | | 2.54 | | |
| Q | | .030 | | 0.76 | 5 |
| TL | .028 | .048 | 0.71 | 1.22 | 3,4 |
| TW | .036 | .046 | 0.91 | 1.17 | |
| r | | .010 | | 0.25 | 3, 10 |
| α | 45° TP | | 45° TP | | 6 |

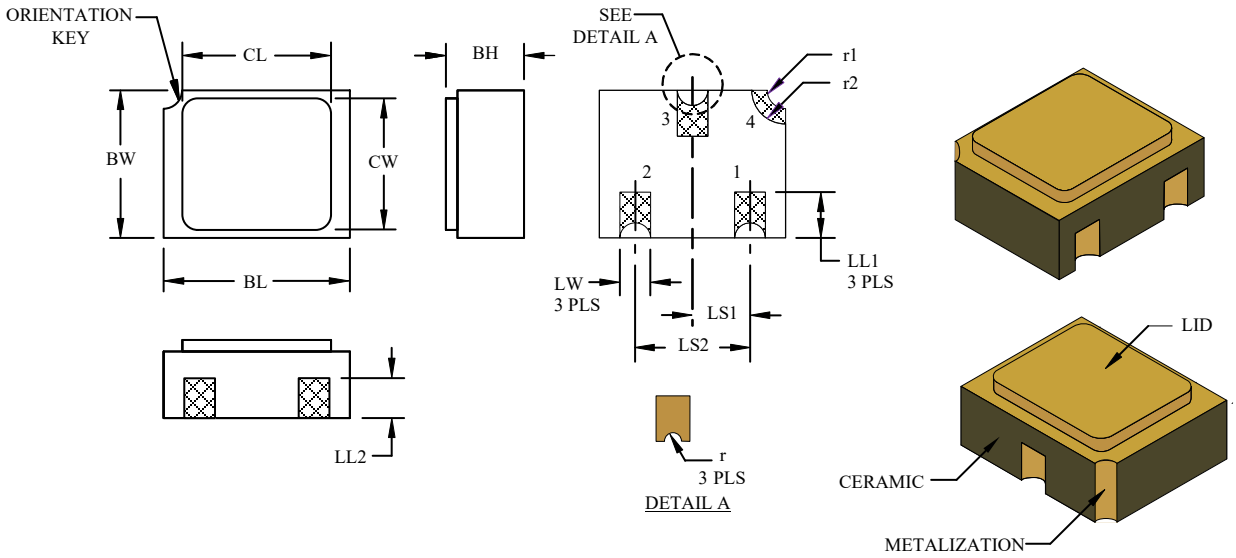



NOTES:
1. Dimension are in inches.
2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TL shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
7. Dimension LU applies between $L_1$ and $L_2$. Dimension LD applies between $L_2$ and LL minimum. Diameter is uncontrolled in $L_1$ and beyond LL minimum.
8. All three leads.
9. The gate shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. Lead 1 = source, lead 2 = drain, lead 3 = gate.

FIGURE 1. Physical dimensions TO-18.



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| Ltr. | Dimensions | | | | Note | Ltr. | Dimensions | | | | Note |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Inches | | Millimeters | | | | Inches | | Millimeters | | |
| | Min | Max | Min | Max | | | Min | Max | Min | Max | |
| BH | .046 | .056 | 1.17 | 1.42 | | LS1 | .035 | .040 | 0.89 | 1.02 | |
| BL | .115 | .128 | 2.92 | 3.25 | | LS2 | .071 | .079 | 1.80 | 2.01 | |
| BW | .085 | .108 | 2.16 | 2.74 | | LW | .016 | .024 | 0.41 | 0.61 | |
| CL | | .128 | | 3.25 | | r | | .008 | | 0.20 | |
| CW | | .108 | | 2.74 | | r1 | | .012 | | 0.31 | |
| LL1 | .022 | .038 | 0.56 | 0.96 | | r2 | | .022 | | 0.56 | |
| LL2 | .017 | .035 | 0.43 | 0.89 | | | | | | | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Hatched areas on package denote metallized areas.
4. Lid material: Kovar.
5. Pad 1 = Drain, Pad 2 = Source, Pad 3 = Gate, Pad 4 = Shielding connected to the lid.

* FIGURE 2. Physical dimensions for surface mount.



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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1 and 2.

3.4.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Radiation hardness assurance (RHA). Radiation hardness assurance requirements, PIN designators, and test levels shall be as defined in MIL-PRF-19500.



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3.6 Electrostatic discharge sensitive (ESDS). The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements of MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

* h. Gate must be terminated to source, R ≤ 100 kΩ, whenever bias voltage is applied drain to source.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.9 Marking. Marking shall be in accordance with MIL-PRF-19500 except for the UB suffix package. Marking on the UB package shall consist of an abbreviated part number, the date code, and the manufacturer's symbol or logo. The prefixes JAN, JANTX, JANTXV, and JANS can be abbreviated as J, JX, JV, and JS respectively.

3.10 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table IV tests, the tests specified in table IV herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.



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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen | Measurements for JANS level | Measurements for JANTX and JANTXV levels |
|---|---|---|
| 9 | Not applicable | Not applicable |
| 10 | Not applicable | Not applicable |
| 11 | $I_{D(off)1}$, $r_{ds(on)}$, $I_{GSS1}$, and $I_{DSS}$ | $I_{D(off)1}$, $r_{ds(on)}$, $I_{GSS1}$, and $I_{DSS}$ |
| 12 | See 4.3.1 | See 4.3.1 |
| 13 | Subgroups 2 and 3 of table I herein;<br>$\Delta I_{DSS}$ = ±15 percent;<br>$\Delta r_{ds(on)}$ = ±20 percent;<br>$\Delta I_{D(off)1}$ = 0.1 nA or ±100 percent of initial value.<br>$\Delta I_{GSS1}$ = ±0.1 nA or ±100 percent of initial value, whichever is greater. | Subgroup 2 of table I herein;<br>$\Delta I_{DSS}$ = ±15 percent;<br>$\Delta r_{ds(on)}$ = ±20 percent;<br>$\Delta I_{D(off)1}$ = 0.1 nA or ±100 percent of initial value.<br>$\Delta I_{GSS1}$ = ±0.1 nA or ±100 percent of initial value, whichever is greater. |

4.3.1 Power burn-in. Power burn-in conditions are in accordance with method 1039 of MIL-STD-750, test condition A and as follows: $T_A \geq +150°C$; $V_{GS}$ = 80 percent of rated, $V_{DS}$ = 0. NOTE: No heatsink or forced air cooling on the devices shall be permitted.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables VIA (JANS) and VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and as follows. Delta measurements shall be in accordance with table II herein.

4.4.2.1 Group B inspection, table E-VIA (JANS) of MIL-PRF-19500.

| | Subgroup | Method | Condition |
|---|---|---|---|
| | B4 | 1037 | $P_T$ = 360 mW at $T_A$ = +30°C, ±5°C; 2,000 cycles. |
| * | B5 | 1027 | 1,000 hours minimum, $T_A$ = 25°C, $V_{DS}$ = 15 V, $I_{DS}$ = 24 mA, or average power = 360 mW or adjusted to achieve $T_J$ = +200°C. |



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4.4.2.2 Group B inspection, table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1027 | $V_{GS}$ = -32 V dc for 2N4856, 2N4857, 2N4858, at $T_A$ = +175°C ±5°C; $V_{DS}$ = 0 V.<br>$V_{GS}$ = -24 V dc for 2N4859, 2N4860, 2N4861, at $T_A$ = +175°C ±5°C; $V_{DS}$ = 0 V. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Delta measurements shall be in accordance with table II herein.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E, (not applicable to UB suffix device). |
| C6 | 1026 | $V_{GS}$ = -32 V dc for 2N4856, 2N4857, 2N4858, at $T_A$ = +175°C ±5°C; $V_{DS}$ = 0 V.<br>$V_{GS}$ = -24 V dc for 2N4859, 2N4860, 2N4861, at $T_A$ = +175°C ±5°C; $V_{DS}$ = 0 V. |

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table IV herein. Delta measurements shall be in accordance with table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.



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TABLE I. Group A inspection

| Inspection 1/ 2/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Gate to source breakdown voltage | 3401 | Bias condition C; $I_G$ = -1.0 µA dc; $V_{DS}$ = 0 | $V_{(BR)GSS}$ | | | |
| 2N4856, 2N4857, 2N4858 | | | | -40 | | V dc |
| 2N4859, 2N4860, 2N4861 | | | | -30 | | V dc |
| Gate reverse current | 3411 | Bias condition C; $V_{DS}$ = 0 | $I_{GSS1}$ | | | |
| 2N4856, 2N4857, 2N4858 | | $V_{GS}$ = -20 V dc | | | -0.25 | nA dc |
| 2N4859, 2N4860, 2N4861 | | $V_{GS}$ = -15 V dc | | | -0.25 | nA dc |
| Drain current cutoff | 3413 | Bias condition A, $V_{DS}$ = 15 V dc, $V_{GS}$ = -10 V dc | $I_{D(off)1}$ | | 0.25 | nA dc |
| Drain current zero-gate voltage | 3413 | Bias condition C, $V_{DS}$ = 15 V dc, $V_{GS}$ = 0, pulsed (see 4.5.1) | $I_{DSS}$ | | | |
| 2N4856, 2N4859 | | | | 50 | 175 | mA dc |
| 2N4857, 2N4860 | | | | 20 | 100 | mA dc |
| 2N4858, 2N4861 | | | | 8 | 80 | mA dc |
| Drain to source on-state voltage | 3405 | Bias condition B, $V_{GS}$ = 0 | $V_{DS(on)}$ | | | |
| 2N4856, 2N4859 | | $I_D$ = 20 mA dc | | | 0.75 | V dc |
| 2N4857, 2N4860 | | $I_D$ = 10 mA dc | | | 0.50 | V dc |
| 2N4858, 2N4861 | | $I_D$ = 5 mA dc | | | 0.50 | V dc |
| Gate to source off-state voltage | 3403 | $V_{DS}$ = 15 V dc, $I_D$ = 0.5 nA dc | $V_{GS(off)}$ | | | |
| 2N4856, 2N4859 | | | | -4 | -10 | V dc |
| 2N4857, 2N4860 | | | | -2 | -6 | V dc |
| 2N4858, 2N4861 | | | | -0.8 | -4 | V dc |
| Static drain to source on-state resistance | 3421 | Condition B, $V_{GS}$ = 0; $I_D$ = 1.0 mA dc; | $r_{ds(on)}$ | | | |
| 2N4856, 2N4859 | | | | | 25 | Ω |
| 2N4857, 2N4860 | | | | | 40 | Ω |
| 2N4858, 2N4861 | | | | | 60 | Ω |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ 2/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +150°C | | | | |
| Gate reverse current | 3411 | Bias condition C; $V_{DS}$ = 0 V | $I_{GSS2}$ | | | |
| 2N4856, 2N4857, 2N4858 | | $V_{GS}$ = -20 V dc | | | -0.5 | µA dc |
| 2N4859, 2N4860, 2N4861 | | $V_{GS}$ = -15 V dc | | | -0.5 | µA dc |
| Drain current | 3413 | Bias condition A, $V_{DS}$ = 15 V dc; $V_{GS}$ = -10 V dc | $I_{D(off)2}$ | | 0.5 | µA dc |
| Subgroup 4 | | | | | | |
| Small-signal common-source short-circuit input capacitance | 3431 | $V_{DS}$ = 0, $V_{GS}$ = -10 V; f = 1 MHz; $C_1$ = .1 µF, $C_2$= 20.1 µF, $L_1$ = $L_2$ = ≥ 500 µH | $C_{iss}$ | | 18 | pF |
| Small-signal common-source short-circuit reverse transfer capacitance | 3433 | $V_{DS}$ = 0, $V_{GS}$ = -10 V; f = 1 MHz; $C_1$ = .1 µF, $L_1$ = $L_2$ = ≥ 500 µH | $C_{rss}$ | | 8 | pF |
| Turn-on delay time | 3459 | See figure 3 | $t_{d(on)}$ | | | |
| 2N4856, 2N4859 | | | | | 6 | ns |
| 2N4857, 2N4860 | | | | | 6 | ns |
| 2N4858, 2N4861 | | | | | 10 | ns |
| Rise time | 3459 | See figure 3 | $t_r$ | | | |
| 2N4856, 2N4859 | | | | | 3 | ns |
| 2N4857, 2N4860 | | | | | 4 | ns |
| 2N4858, 2N4861 | | | | | 10 | ns |
| Turn-off delay time | 3459 | See figure 3 | $t_{d(off)}$ | | | |
| 2N4856, 2N4859 | | | | | 25 | ns |
| 2N4857, 2N4860 | | | | | 50 | ns |
| 2N4858, 2N4861 | | | | | 100 | ns |
| Subgroups 5, 6, and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-PRF-19500.
2/ These characteristics applicable to all package styles, unless otherwise noted.



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TABLE II. Groups B, C, and E delta measurements. 1/ 2/ 3/ 4/

| Step | Inspection | MIL-STD-750 | | Symbol | Limit | Unit |
|---|---|---|---|---|---|---|
| | | Method | Conditions | | | |
| 1. | Static drain to source on-state resistance | 3421 | Condition B, $V_{GS}$ = 0; $I_{DS}$ = 1.0 mA dc; | $\Delta r_{DS(on)}$ | ±25 percent change from previously measured value. | |
| 2. | Drain current | 3413 | Bias condition C; $V_{DS}$ = 15 V dc, $V_{GS}$ = 0; pulsed (see 4.5.1) | $\Delta I_{DSS1}$ | ±15 percent change from previously measured value. | |

1/ Devices which exceed the group A limits for this test shall be rejected.
2/ The delta measurements for table E-VIA (JANS) of MIL-PRF-19500 are as follows: Subgroup 5, see table II herein, steps 1 and 2.
3/ The delta measurements for table E-VII of MIL-PRF-19500 are as follows: Subgroup 6, table II herein, steps 1 and 2 (for JANS only).
4/ The delta measurements for table E-IX of MIL-PRF-19500 are as follows: Subgroups 1 and 2, table II herein, steps 1 and 2.



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TABLE III Group D inspection.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 2 | | $T_C$ = +25°C | | | | |
| * Steady-state total dose Irradiation ($V_{GS}$ bias) 4/ | 1019 | Condition A, $V_{GS}$ = -32 V, $V_{DS}$ = 0 | | | | |
| * Steady-state total dose Irradiation ($V_{DS}$ bias) 4/ | 1019 | Condition A, $V_{GS}$ = 0, $V_{DS}$ = 80 percent of rated $V_{DS}$ (pre-irradiation) | | | | |
| Gate to source breakdown voltage | 3401 | Bias condition C; $I_G$ = -1.0 µA dc; $V_{DS}$ = 0 | $V_{(BR)GSS}$ | | | |
| 2N4856, 2N4857, 2N4858 | | | | -40 | | V dc |
| 2N4859, 2N4860, 2N4861 | | | | -30 | | V dc |
| Gate reverse current | 3411 | Bias condition C; $V_{DS}$ = 0 | $I_{GSS1}$ | | | |
| 2N4856, 2N4857, 2N4858 | | $V_{GS}$ = -20 V dc | | | -0. 5 | nA dc |
| 2N4859, 2N4860, 2N4861 | | $V_{GS}$ = -15 V dc | | | -0.5 | nA dc |
| Drain current cutoff | 3413 | Bias condition A, $V_{DS}$ = 15 V dc, $V_{GS}$ = -10 V dc | $I_{D(off)1}$ | | 2.5 | nA dc |
| Drain current zero-gate voltage | 3413 | Bias condition C, $V_{DS}$ = 15 V dc, $V_{GS}$ = 0, pulsed (see 4.5.1) | $I_{DSS}$ | | | |
| 2N4856, 2N4859 | | | | 50 | 175 | mA dc |
| 2N4857, 2N4860 | | | | 20 | 100 | mA dc |
| 2N4858, 2N4861 | | | | 8 | 80 | mA dc |
| Drain to source on-state voltage | 3405 | Bias condition B, $V_{GS}$ = 0 | $V_{DS(on)}$ | | | |
| 2N4856, 2N4859 | | $I_D$ = 20 mA dc | | | 0.75 | V dc |
| 2N4857, 2N4860 | | $I_D$ = 10 mA dc | | | 0.50 | V dc |
| 2N4858, 2N4861 | | $I_D$ = 5 mA dc | | | 0.50 | V dc |
| Gate to source off-state voltage | 3403 | $V_{DS}$ = 15 V dc, $I_D$ = 0.5 nA dc | $V_{GS(off)}$ | | | |
| 2N4856, 2N4859 | | | | -4 | -10 | V dc |
| 2N4857, 2N4860 | | | | -2 | -6 | V dc |
| 2N4858, 2N4861 | | | | -0.8 | -4 | V dc |
| Static drain to source on-state resistance | 3421 | Condition B, $V_{GS}$ = 0; $I_D$ = 1.0 mA dc | $r_{ds(on)}$ | | | |
| 2N4856, 2N4859 | | | | | 25 | Ω |
| 2N4857, 2N4860 | | | | | 40 | Ω |
| 2N4858, 2N4861 | | | | | 60 | Ω |

1/ Tests to be performed on all devices receiving radiation exposure.
2/ For sampling plan, see MIL-PRF-19500.
3/ Electrical characteristics apply to the corresponding non suffix versions and UB suffix unless otherwise noted.
4/ Separate samples shall be pulled for each bias.



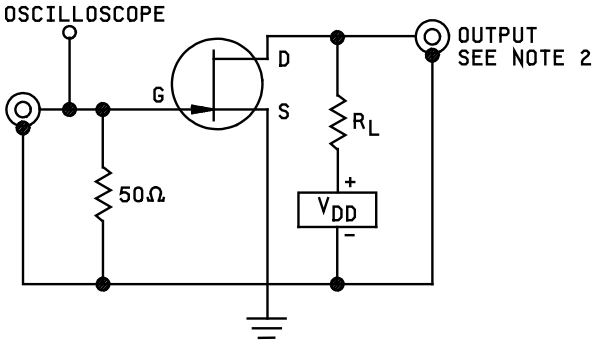
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TABLE IV. Group E inspection (all quality levels) for qualification or re-qualification only.

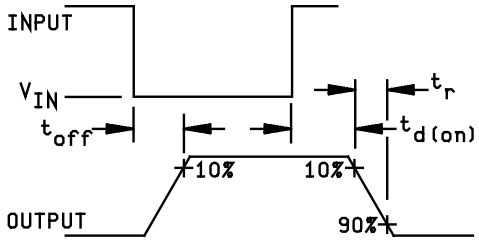
| Inspection | MIL-STD-750 | | Sample plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices<br>c = 0 |
| Temperature cycling | 1051 | 500 cycles. | |
| Hermetic seal<br>Fine leak<br>Gross leak | 1071 | | |
| End-point electrical measurements | | See table I, subgroup 2 and table II herein. | |
| Subgroup 2 | | | 45 devices<br>c = 0 |
| Blocking life | 1048 | Test temperature = +125°C, $V_{GS}$ or $V_{DG}$ = 80 percent of rated, T = 1,000 hours. | |
| End-point electrical measurements | | See table I, subgroup 2 and table II herein | |
| Subgroup 4 and 5 | | | |
| Not applicable | | | |



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TEST CIRCUIT




VOLTAGE WAVEFORMS

| TEST CONDITIONS AND COMPONENT VALUE | | | | | | |
|---|---|---|---|---|---|---|
| Type | $V_{DD}$ | $V_{GS(on)}$ | $V_{GS(off)}$ | $R_L$ | $V_{IN}$ | $I_{D(on)}$ (1) |
| | V dc | V dc | V dc | Ω | V dc | mA dc |
| 2N4856, 2N4859 | 10 | 0 | -10 | 464 | -10 | 20 |
| 2N4857, 2N4860 | 10 | 0 | -6 | 953 | -6 | 10 |
| 2N4858, 2N4861 | 10 | 0 | -4 | 1,910 | -4 | 5 |

(1) Nominal value; exact value varies slightly with transistor parameters.

NOTES:

1. The input waveform has the following characteristics: $t_p$ = 200 ns; $t_r \leq 1$ ns; duty cycle ≈ 2 percent. It is supplied by a generator with $Z_{out}$ = 50 Ω.
2. Waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 0.75$ ns; $R_{IN} \geq 1$ MΩ; $C_{IN} \leq 2.5$ pF.
3. These characteristics applicable to all package styles, unless otherwise noted.

FIGURE 3. Switching time test circuit.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

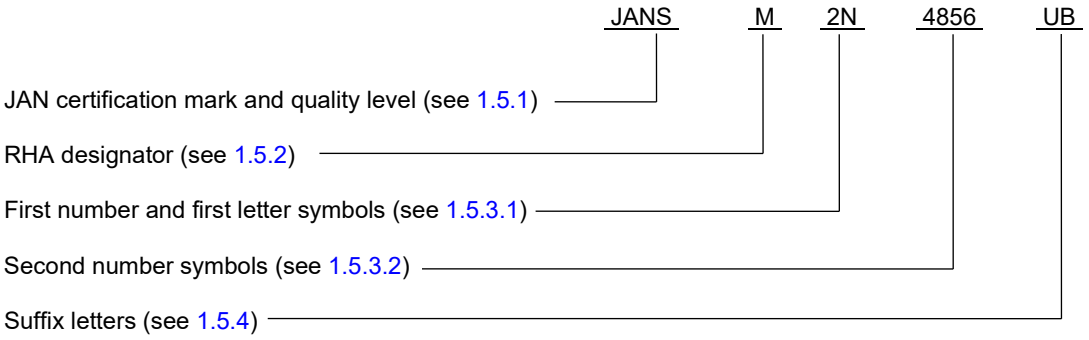
d. The complete Part or Identifying Number (PIN), see 1.5 and 6.5.

e. For acquisition of RHA designated devices, table III, subgroup 1 testing of group D herein is optional. If subgroup 1 is desired, it should be specified in the contract or order.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example.

6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.






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6.5 List of PINs.

6.5.1 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

| PINs for devices of the base quality level | PINs for devices of the "TX" quality level | PINs for devices of the "TXV" quality level | PINs for devices of the "S" quality level |
|---|---|---|---|
| JAN2N4856 | JANTX2N4856 | JANTXV2N4856 | JANS#2N4856 |
| JAN2N4856UB | JANTX2N4856UB | JANTXV2N4856UB | JANS#2N4856UB |
| JAN2N4857 | JANTX2N4857 | JANTXV2N4857 | JANS#2N4857 |
| JAN2N4857UB | JANTX2N4857UB | JANTXV2N4857UB | JANS#2N4857UB |
| JAN2N4858 | JANTX2N4858 | JANTXV2N4858 | JANS#2N4858 |
| JAN2N4858UB | JANTX2N4858UB | JANTXV2N4858UB | JANS#2N4858UB |
| JAN2N4859 | JANTX2N4859 | JANTXV2N4859 | JANS#2N4859 |
| JAN2N4859UB | JANTX2N4859UB | JANTXV2N4859UB | JANS#2N4859UB |
| JAN2N4860 | JANTX2N4860 | JANTXV2N4860 | JANS#2N4860 |
| JAN2N4860UB | JANTX2N4860UB | JANTXV2N4860UB | JANS#2N4860UB |
| JAN2N4861 | JANTX2N4861 | JANTXV2N4861 | JANS#2N4861 |
| JAN2N4861UB | JANTX2N4861UB | JANTXV2N4861UB | JANS#2N4861UB |

(1) The number sign (#) represent one of eight RHA designators available (M, D, P, L, R, F, G, or H). The PIN is also available without a RHA designator.

* 6.6 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
* Navy - SH
Air Force - 85
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Review activities:
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